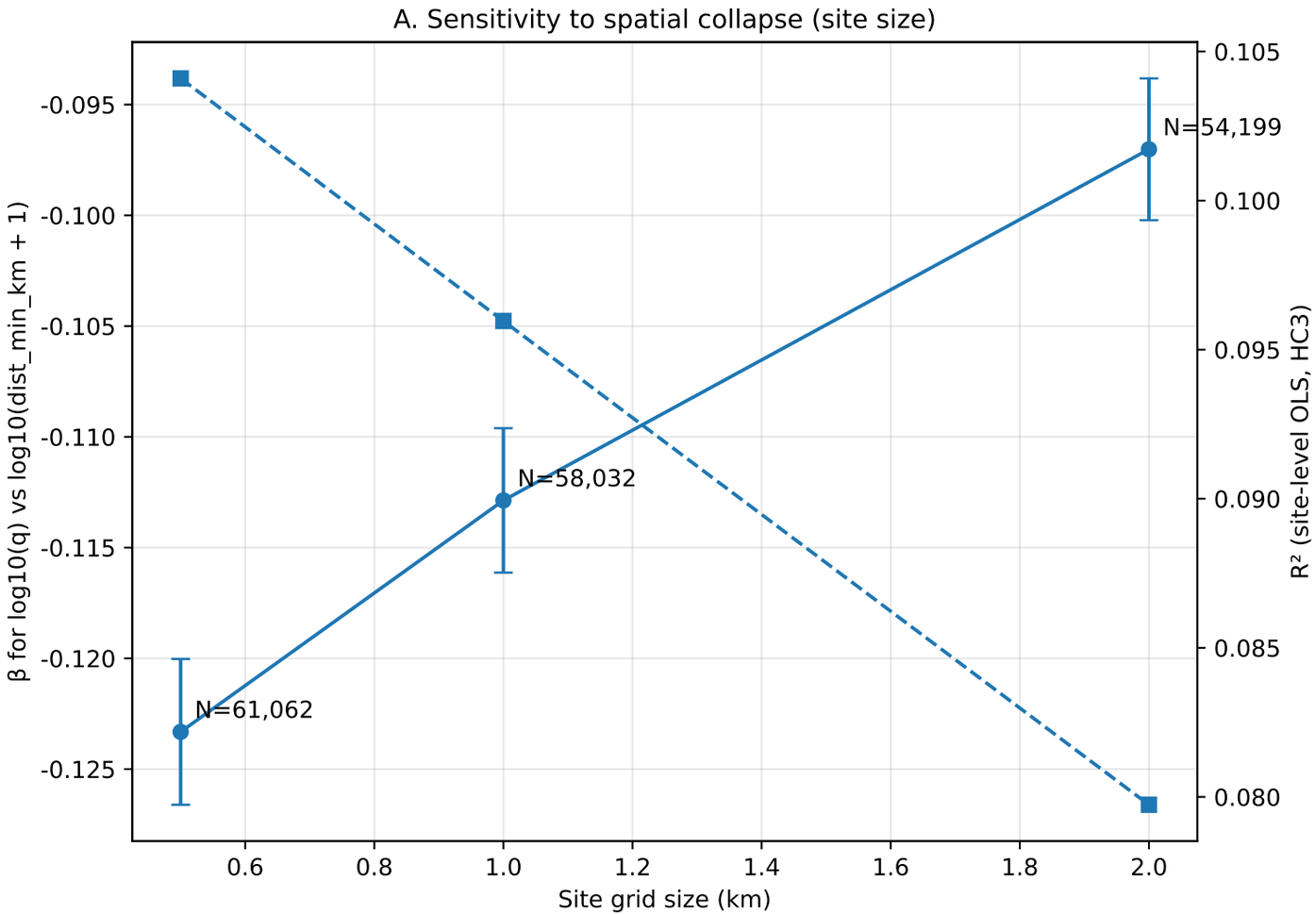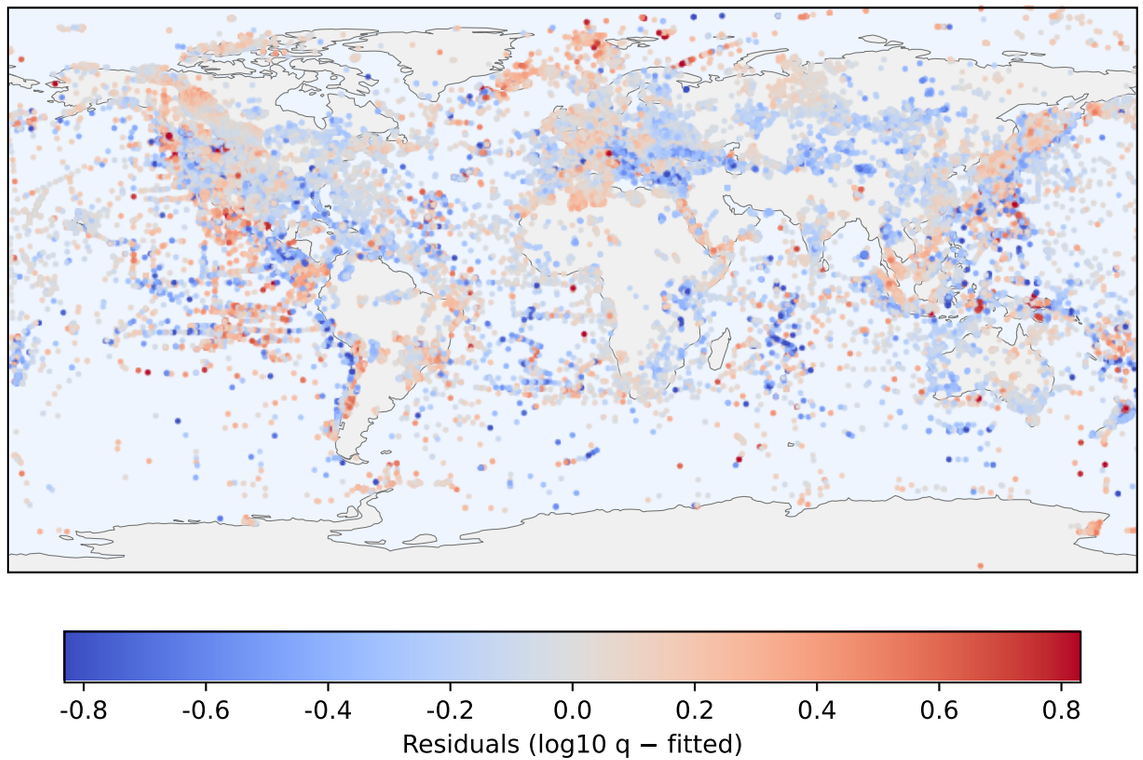Density-independence diagnostic: site-size sensitivity and residual structure